UMKC Impact & Translation Series

Making Sense of Technology Readiness Levels

A Practical Framework for Moving Your Work Toward Real-World Impact

Technology Readiness Levels (TRLs) provide a structured, universal framework to assess the maturity of innovations as they move from initial discovery to market adoption. Originally developed by NASA, TRLs are now used across federal agencies and industries to evaluate readiness and mitigate risks.

For UMKC researchers, understanding TRLs offers a strategic advantage in project planning, commercialization, and industry engagement. This whitepaper introduces the TRL framework and explores how it applies across diverse academic disciplines, from medicine to the arts.

This guide is not intended to make you an expert in Technology Readiness Levels. It is intended to help you use TRLs as a practical tool to better understand where your work stands today, what risks remain, and what would need to happen for that work to function outside the university.

UITS Whitepaper

This whitepaper is part of the **UMKC Impact & Translation Series (UITS)** - a set of short videos and supporting materials designed to help faculty think more clearly about how their work can move beyond the university and into real-world use.

The videos introduce core ideas. This document goes deeper on one of those ideas: **how to use TRLs to assess the readiness of your work for real-world use.**

Why TRLs Matter at UMKC

At UMKC, faculty members contribute transformative ideas across a wide array of disciplines. Whether developing life-saving therapies, advancing next-generation technologies, or creating groundbreaking artistic works, researchers often face the same question: "How ready is this innovation for real-world application?"

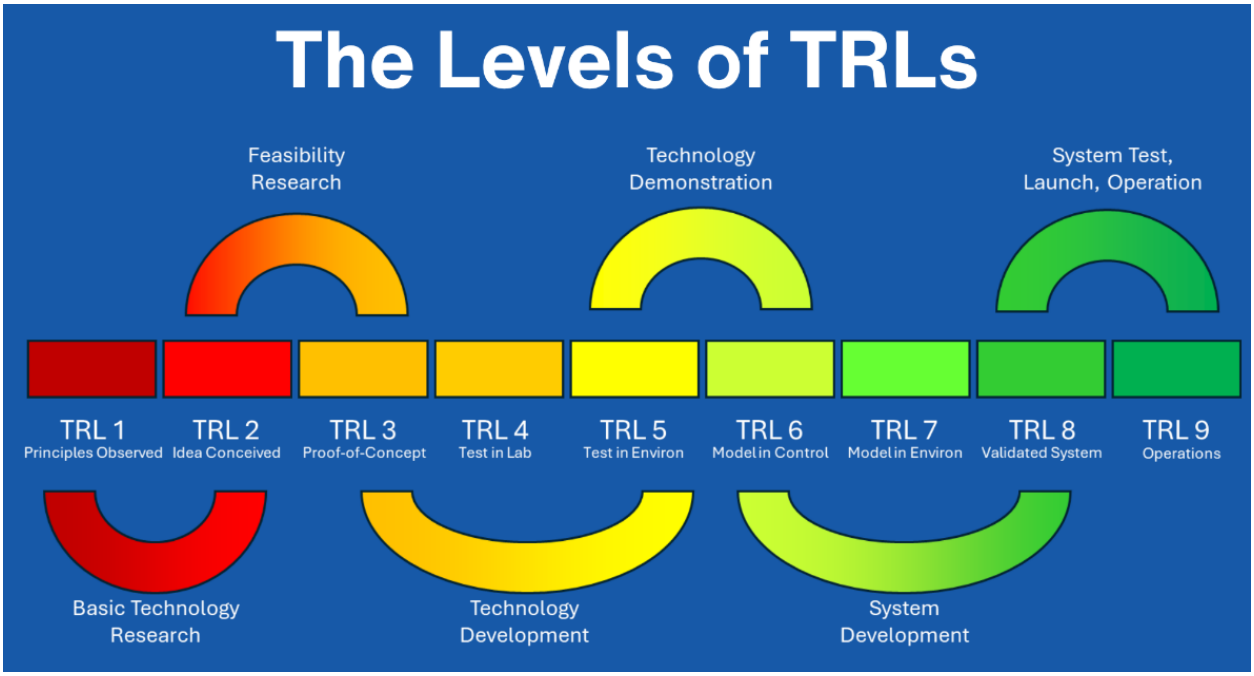
Technology Readiness Levels offer a clear answer. These nine levels, which mark the journey from concept to full deployment, allow innovators to communicate maturity levels with funders, partners, and technology transfer offices.

Importantly, TRLs also help identify necessary steps to advance a project—whether that means prototyping, clinical testing, or regulatory navigation. Understanding this framework positions UMKC researchers to more effectively translate discoveries into societal impact.

Within the UMKC Impact & Translation Series, TRLs serve as one way to think about readiness. They do not measure how good or innovative an idea is. They help clarify how much uncertainty has been reduced and how close that work is to functioning in a real-world setting.

Understanding the TRL Scale

TRLs offer a shared vocabulary to describe the evolution of an idea into a usable, validated solution. Each level marks a critical milestone that builds upon the last, creating a roadmap from research to deployment.



TRL 1: Foundational Discovery - Basic principles have been observed. This level includes early-stage research and published findings that form the basis for a new idea.

TRL 2: Concept Formulation - Practical applications are identified, but remain speculative. Initial hypotheses and theoretical models begin to take shape.

TRL 3: Experimental Proof of Concept - Laboratory studies validate core assumptions. Experiments demonstrate that the concept works under controlled conditions.

TRL 4: Lab-Based Validation - Key components are built and tested in a laboratory setting. The system or process shows functionality and feasibility in a simplified environment.

TRL 5: Simulated Environment Testing - Prototypes or components are validated in environments that mimic real-world conditions. This marks the beginning of bridging lab and field.

TRL 6: Relevant Environment Demonstration - A system or prototype is tested in a setting that reflects real operating conditions. This is typically the final step before operational field testing.

TRL 7: Operational Environment Demonstration - The prototype is deployed in an actual operating environment. The system is integrated with real-world infrastructure or workflows.

TRL 8: Final System Qualification - The complete system undergoes rigorous testing and quality assurance to meet all operational criteria. Final refinements are made.

TRL 9: Proven System in Use - The technology is in full use and has proven effective through successful mission operations or market validation.

Most academic research begins in TRL 1–3. Real-world use typically requires reaching TRL 6–9. Each of these levels provides a critical opportunity for UMKC researchers to assess progress, plan next steps, and communicate readiness to stakeholders.

How TRLs Apply Across UMKC Research Fields

TRLs are not just for aerospace and engineering. Their structured approach to development applies across UMKC's diverse research landscape.

TRLs focus on technical development, but technical readiness alone is not enough to create impact. Work may function in a lab or controlled setting and still fail to translate if it does not align with real-world conditions, users, or constraints.

As you assess your work, it is important to consider both:

- Technical readiness (Does it work?)
- Real-world readiness (Does it matter? Will it be used?)

Strong translation requires progress on both dimensions. Understanding how TRLs map to your field helps clarify goals, timelines, and partnership opportunities.

Drug Development: From Molecules to Medicine

Translating laboratory discoveries into safe, effective therapies is a long and complex process. Drug development involves rigorous evaluation to ensure both safety and efficacy before a compound ever reaches a patient.

The TRL framework maps this journey clearly—from understanding molecular mechanisms to navigating human trials and regulatory approvals—helping researchers anticipate the milestones and requirements needed for each stage.

Basic Research (TRL 1-3) - Exploration of disease mechanisms and initial compound screening define early progress in pharmaceuticals. This work establishes theoretical and empirical groundwork.

Preclinical Testing (TRL 4-5) – Candidate drugs undergo lab-based and animal studies to assess safety and efficacy. Validation here supports decisions to move into human trials.

Clinical Trials (TRL 6-7) - Phase I and II trials test the drug in humans, assessing dosage, effectiveness, and side effects. Success indicates real-world feasibility.

Market Readiness (TRL 8-9) - Phase III trials and FDA approval represent system qualification. Once the drug is prescribed and in use, it reaches full TRL 9 maturity.

Medical Devices: Engineering with Impact

Medical devices are at the intersection of engineering, health sciences, and regulatory oversight. Whether developing wearable sensors, imaging systems, or surgical tools, innovators must demonstrate both technical performance and patient safety.

TRLs provide a structured lens through which UMKC researchers can advance from a promising concept to a clinically accepted tool, highlighting the steps from benchtop validation to human trials and full deployment in care settings.

Concept and Feasibility (TRL 1-3) - Innovators explore materials, mechanisms, and user needs. Early-stage prototypes and simulations guide the design.

Bench Testing (TRL 4-6) - Devices are tested in labs and simulated environments. FDA design controls often begin during this phase to prepare for regulatory pathways.

Human Trials and Approvals (TRL 7-9) - Clinical trials, 510(k) clearance, or PMA approval bring the technology into clinical practice, validating safety and effectiveness.

Engineering Projects: Building Toward Real-World Solutions

Engineering innovation often begins with theoretical models and progresses through iterative design, testing, and deployment. Whether focused on infrastructure, energy systems, or digital platforms, projects must evolve from feasibility to function in real-world environments. TRLs help engineering researchers at UMKC chart a logical path from initial sketches to robust, field-tested solutions—ensuring each stage of development meets increasing levels of rigor and relevance.

Design and Modeling (TRL 1-3) - Engineers use sketches, CAD models, and theoretical analyses to explore a concept's feasibility and potential.

Prototype Development (TRL 4-6) - Physical builds and lab-based performance testing are conducted. Teams begin integrating components into working systems.

Deployment and Optimization (TRL 7-9) - Solutions are tested in the field—on roadways, buildings, or data networks. Refinements occur through real-world performance feedback.

Creative and Digital Works: Innovation in Expression

Creative works and digital experiences follow their own arc of development, from inspiration and experimentation to release and audience engagement.

For artists, designers, and digital developers, TRLs offer a way to frame the creative process within a development pathway—recognizing the maturation of ideas through prototyping, refinement, and real-world interaction. This perspective empowers UMKC creators to plan, iterate, and launch with greater strategic clarity.

Ideation and Mockups (TRL 1-3) - Artists, designers, and developers explore themes and create initial drafts, whether musical compositions, visual concepts, or code frameworks.

Prototype and Beta Testing (TRL 4-6) - Works take form in testable formats such as demo recordings, app betas, or interactive installations.

Public Release (TRL 7-9) - Distribution, exhibition, or public use validates impact and readiness. When a digital tool or artistic work is actively engaged with by its intended audience, it reaches maturity.

This cross-sector versatility makes TRLs an invaluable planning and communication tool for UMKC's research community.

Integrating TRLs with Manufacturing Readiness

Once technologies reach the mid-to-late TRLs, questions of scale, consistency, and supply chain reliability come into focus. That's where Manufacturing Readiness Levels (MRLs) align with TRLs to guide decisions around production.

Prototype Manufacturing (MRL 4-6) - Processes are developed to produce small batches of the innovation. Key challenges such as quality assurance and repeatability are addressed.

Full Production Capability (MRL 7-9) - Systems are in place for large-scale manufacturing, including process controls, workforce training, and supplier management. This stage aligns closely with TRL 8-9, ensuring the technology is not only usable but also reproducible at scale.

In practice, TRLs and MRLs often advance together as work moves closer to real-world use. Ignoring manufacturing readiness too long can delay or prevent impact. UMKC researchers working on products with commercial potential should begin MRL planning by TRL 5-6 to align timelines and resources effectively.

Planning Next Steps Using TRLs at UMKC

TRLs are most valuable when used to guide decision-making and project development. At UMKC, researchers can leverage the framework to clarify needs and unlock resources.

Guiding Question:

Where does your work currently stand in terms of readiness, and what would need to happen for it to function outside the university?

1. **Self-Assess Your Project** - Identify your current TRL and what evidence supports that assessment. Then define what would need to be true to move to the next level.

2. **Meet with the Office of Commercialization** - Use TRLs as a starting point for conversation, not a conclusion. UMKC's commercialization staff can help verify your TRL stage, identify funding sources, and connect you with development partners or mentors.

3. **Incorporate TRLs into Grant Proposals** - Federal agencies increasingly require a TRL assessment in SBIR, STTR, and translational grant applications. Including this framework signals preparedness.

4. **Leverage Peer Learning Opportunities** - Join faculty learning communities or innovation lunches to workshop TRL strategies and see how others at UMKC have navigated their journey.

The goal is not to assign a number to your work. The goal is to clarify what needs to happen next. By consistently applying this model, researchers can more confidently navigate the path to impact.

Conclusion: A Shared Framework for Innovation Success

Technology Readiness Levels provide UMKC researchers with a structured lens through which to view their innovation journey. From initial insight to public benefit, TRLs clarify both where you are and what must come next.

Importantly, this model transcends disciplines and makes it easier for faculty to align with industry, investors, and funders.

Technology Readiness Levels are most useful when they change how you think about your work. They provide a way to step back, assess what has been proven, identify what remains uncertain, and understand what it would take for your work to function beyond the university. That clarity is what enables impact.

Resources & Contacts

UMKC Office of Commercialization
Director: Jeff Blackwood, MBA
📍 Website: www.umkcinnovates.com/tech-commercialization
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- NASA Technology Readiness Level Guidebook
- FDA Drug Development Process
- Office of the Under Secretary of Defense TRA Guide